Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

- **Official Study Title:** A Randomized, Open-label, Phase 3 Study of MK-2870 in Combination With Pembrolizumab Compared to Pembrolizumab Monotherapy in the First-line Treatment of Participants With Metastatic Non-small Cell Lung Cancer With PD-L1 TPS Greater Than or Equal to 50%
- **Registry/Brief Title:** Sacituzumab Tirumotecan (MK-2870) in Combination With Pembrolizumab Versus Pembrolizumab Alone in Metastatic Non-small Cell Lung Cancer (NSCLC) With Programmed Cell Death Ligand 1 (PD-L1) Tumor Proportion Score (TPS) ≥ 50% (MK-2870-007)
- **Short Title/Acronym:** TroFuse-007
- **Protocol Number:** MK-2870-007
- **NCT Number:** NCT06170788
- **Trial Status:** RECRUITING (Currently recruiting participants)
- **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC)
- **Investigational Product:** Sacituzumab Tirumotecan (MK-2870) and Pembrolizumab
- **Phase of Development:** Phase 3
- **Medical Monitor:** [Sponsor Medical Director / Merck Contact]

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To compare sacituzumab tirumotecan (MK-2870) combined with pembrolizumab to pembrolizumab alone with respect to overall survival (OS). The primary hypothesis is that the combination of sacituzumab tirumotecan and pembrolizumab is superior to pembrolizumab alone with respect to OS. * **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Objective Response (OR) / Objective Response Rate (ORR), Duration of Response (DOR) by Blinded Independent Central Review (BICR) per RECIST 1.1, patient-reported Quality of Life (EORTC QLQ-C30), and the overall safety and tolerability of the combination regimen.

2.2 Background & Rationale To evaluate whether combining a TROP2-directed antibody-drug conjugate (sacituzumab tirumotecan/ MK-2870) with an immune checkpoint inhibitor (pembrolizumab) provides superior overall survival compared to pembrolizumab monotherapy. Pembrolizumab alone is currently a standard of care for the first-line treatment of patients with metastatic NSCLC and high PD-L1 expression (TPS $\ge 50\%$). The study aims to determine if the addition of MK-2870 prolongs cancer-free periods and improves survival outcomes.

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Pembrolizumab monotherapy) * **Blinding:** Open-label * **Randomization:** 1:1 Randomized

3.2 Planned Enrollment & Duration * **Target N:** Approximately 614 participants * **Number of Sites:** Multicenter / Global * **Estimated Study Start:** 15-Dec-2023 * **Estimated Primary Completion:** 25-Jan-2028

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults ≥ 18 years of age. 2. Histologically or cytologically confirmed diagnosis of advanced/ metastatic squamous or nonsquamous non-small cell lung cancer (NSCLC). 3. Confirmation that epidermal growth factor receptor- (EGFR-), anaplastic lymphoma kinase- (ALK-), or proto-oncogene tyrosine-protein kinase ROS (ROS1-) directed therapy is not indicated as primary therapy. 4. Provided tumor tissue that demonstrates programmed cell death ligand 1 (PD-L1) expression in $\ge 50\%$ of tumor cells as assessed by an immunohistochemistry (IHC) central laboratory. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 assessed within 7 days before randomization. 6. A life expectancy of at least 3 months. 7. Human immunodeficiency virus (HIV)-infected participants must have well controlled HIV on antiretroviral therapy (ART).

4.2 Exclusion Criteria 1. Diagnosis of small cell lung cancer or, for mixed tumors, presence of small cell elements. 2. Has Grade ≥ 2 peripheral neuropathy. 3. History of documented severe dry eye syndrome, severe Meibomian gland disease and/or blepharitis, or severe corneal disease that prevents/delays corneal healing. 4. Has active inflammatory bowel disease requiring immunosuppressive medication or previous clear history of inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis, or chronic diarrhea). 5. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease within the 6 months preceding study intervention. 6. Received prior systemic anticancer therapy for their metastatic NSCLC. 7. Received prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent, or with an agent directed to another stimulatory or coinhibitory T-cell receptor (Note: Prior treatment in the neoadjuvant or adjuvant setting for nonmetastatic resectable NSCLC is allowed as long as therapy was completed at least 12 months before diagnosis of metastatic NSCLC). 8. Received prior systemic anticancer therapy including investigational agents within 4 weeks before randomization. 9. Received radiation therapy to the lung that is >30 Gy within 6 months of start of study intervention. 10. Received prior radiotherapy within 2 weeks of start of study intervention, or radiation-related toxicities, requiring corticosteroids. 11. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. Administration of killed vaccines is allowed. 12. Diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy. 13. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 14. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 15. Known intolerance to sacituzumab tirumotecan or pembrolizumab and/or any of their excipients; for pembrolizumab, severe hypersensitivity ($\ge$ Grade 3) is exclusionary. Known hypersensitivity to sacituzumab tirumotecan or other biologic therapy. 16. Active autoimmune disease that has required systemic treatment in the past 2 years. 17. History of (noninfectious) pneumonitis/interstitial lung disease (ILD) that required steroids or has current pneumonitis/ILD. 18. Active infection requiring systemic therapy. 19. Concurrent active Hepatitis B and Hepatitis C virus infection. 20. Human immunodeficiency virus (HIV)-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 21. History of allogeneic tissue/solid organ transplant. 22. Requires treatment with a strong inhibitor or inducer of Cytochrome P450 3A4 (CYP3A4) at least 14 days before the first dose of study intervention and throughout the study.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** * *Experimental Arm:* Sacituzumab tirumotecan via IV infusion on Days 1, 15, and 29 of each 6-week cycle + Pembrolizumab 400 mg via IV infusion on Day 1 of each 6-week cycle. * *Comparator Arm:* Pembrolizumab 400 mg via IV infusion on Day 1 of each 6-week cycle. * **Drug Identifiers:** Pembrolizumab (Trade Name: **Keytruda** | ATC Code: **L01FF02** | EMA Product Information) * **Route of Administration:** Intravenous (IV) infusion. * **Duration of Treatment:** Pembrolizumab for up to 18 cycles (approximately 2 years); MK-2870 until disease progression or unacceptable toxicity. *Note: All participants who have completed the first course of pembrolizumab may be eligible for up to an additional 9 cycles of pembrolizumab monotherapy if there is blinded independent central review (BICR)-verified progressive disease by RECIST 1.1 after initial treatment.*

5.2 Concomitant & Prohibited Therapy * **Permitted:** Supportive care measures, such as diphenhydramine (or equivalent) and H2 antagonists for infusion reactions. * **Prohibited:** Strong CYP3A4 inhibitors/inducers, live or live-attenuated vaccines within 30 days before the first dose, and concurrent investigational anticancer agents.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor tissue PD-L1 testing, Baseline Imaging | | **Baseline** | Day 0 | Randomization, First Dose Administration | | **Interim** | Every 6 Weeks | Safety Labs, Treatment Infusions, Efficacy Assessment (Imaging), QoL Questionnaires (EORTC QLQ-C30) | | **End of Study** | Up to 6 Years | Final Vital Signs, Exit Interview, Survival and Progression Follow-up |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Overall Survival (OS) - defined as the time from randomization to death from any cause. * **Measurement Method:** Clinical tracking and survival analysis.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) by BICR per RECIST 1.1. * Objective Response (OR) / Objective Response Rate (ORR) by BICR per RECIST 1.1. * Duration of Response (DOR) by BICR per RECIST 1.1. * Change from baseline in health-related quality of life via EORTC QLQ-C30. * Safety and tolerability (percentage of participants experiencing AEs).

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard continuous AE tracking, graded by severity. Percentage of participants experiencing AEs and discontinuations due to AEs will be recorded.
- **Serious Adverse Events (SAEs):** Reported continuously per standard oncology trial safety regulations.
- **Data Monitoring Committee (DMC):** An external group of experts (External Data Monitoring Committee) will oversee overall study risk, benefit, and safety.

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy analyses (OS, PFS); Safety Set for patients receiving at least one dose of study medication.
- **Sample Size Justification:** Target Enrollment = 614 participants to provide adequate power to detect a statistically significant improvement in OS for the combination arm.
- **Handling of Missing Data:** Standard survival analysis techniques including Kaplan-Meier estimations and appropriate censoring rules for time-to-event endpoints (e.g., OS, PFS, DOR).

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Regular sponsor-directed onsite and remote monitoring of trial sites.
- **Audit Trail:** Secure eCRF systems, data clarification processes, and strict BICR protocols for imaging to prevent bias.

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11. Study Identifiers & Governance

- **Posting ID:** 975840486149636360
- **Primary ID:** MK-2870-007
- **Registry Number:** NCT06170788
- **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC)
- **Study Title:** TroFuse-007
- **Official Regulatory Title:** A Randomized, Open-label, Phase 3 Study of MK-2870 in Combination With Pembrolizumab Compared to Pembrolizumab Monotherapy in the First-line Treatment of Participants With Metastatic Non-small Cell Lung Cancer With PD-L1 TPS Greater Than or Equal to 50%

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12. Clinical Framework (NSCLC Specific)

- **Disease Area:** Non-small Cell Lung Cancer (NSCLC)
- **Preferred UMLS Name:** Metastatic Non-Small Cell Lung Carcinoma
- **Clinical Classifications:**
    - **MeSH Code:** D002275
    - **SNOMED ID:** 254635002
    - **SNOMED Hierarchy:** 254637007|Malignant neoplasm of lung|Malignant neoplasm of lung (disorder)
- **Diagnosis Threshold:** Programmed Cell Death Ligand 1 (PD-L1) Tumor Proportion Score (TPS) $\ge 50\%$. *Note: PD-L1 assessments utilize Non-Small Cell Lung Cancer (NSCLC) Diagnostic Reagents (Semantic Type: Indicator, Reagent, or Diagnostic Aid).*
- **Medical Methodology:** First-line systemic cancer immunotherapy and targeted Antibody-Drug Conjugate (ADC) therapy.
- **Optimization Target:** Improved Overall Survival (OS) and delayed disease progression.

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13. Study Procedures & Methodology

- **Management Pathway:** Intravenous antineoplastic therapy following standard oncology pathways.

- **Education Protocol (HCP):** Site initiation visits and regular protocol safety training.
- **Education Protocol (Patient):** Informed consent detailing potential treatment risks, AE symptom reporting, and required biopsy/imaging procedures.
- **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all tumor response assessments (RECIST 1.1).

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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 6 years per patient.
- **Onsite Visit Cadence:** Day 1, 15, and 29 of each 6-week cycle for the combination arm; Day 1 of each 6-week cycle for the comparator arm.
- **Imaging/Monitoring Frequency:** Tumor imaging every 6 weeks (or per protocol specifics) until disease progression.
- **Checklist Review Interval:** Routine safety and QoL evaluations at dosing visits.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]